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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 43.2

$$\begin{aligned}
 I &= \int \frac{x^2 + 7x + 1}{(x+2)^2(2x+1)} dx \\
 \frac{x^2 + 7x + 1}{(x+2)^2(2x+1)} &= \frac{A}{(x+2)^2} + \frac{B}{x+2} + \frac{C}{2x+1} \\
 A &= \frac{x^2 + 7x + 1}{2x+1} \Big|_{x=-2} = \frac{4 - 14 + 1}{-4 + 1} = \frac{-9}{-3} = 3 \\
 \frac{x^2 + 7x + 1}{(x+2)^2(2x+1)} &= \frac{3}{(x+2)^2} + \frac{B}{x+2} + \frac{C}{2x+1} \\
 \frac{x^2 + 7x + 1}{(x+2)^2(2x+1)} - \frac{3}{(x+2)^2} &= \frac{x^2 + 7x + 1 - 3(2x+1)}{(x+2)^2(2x+1)} \\
 \frac{x-1}{(x+2)(2x+1)} &= \frac{B}{x+2} + \frac{C}{2x+1} \\
 B &= \frac{x-1}{2x+1} \Big|_{x=-2} = \frac{-3}{-3} = 1 \\
 C &= \frac{x-1}{x+2} \Big|_{x=-\frac{1}{2}} = \frac{-\frac{3}{2}}{\frac{3}{2}} = -1 \\
 \frac{x^2 + 7x + 1}{(x+2)^2(2x+1)} &= \frac{3}{(x+2)^2} + \frac{1}{x+2} - \frac{1}{2x+1} \\
 I &= 3 \int \frac{dx}{(x+2)^2} + \int \frac{dx}{x+2} - \int \frac{dx}{2x+1} \\
 I &= -\frac{3}{x+2} + \ln|x+2| - \frac{1}{2} \ln|2x+1| + C
 \end{aligned}$$

References